

that heat-generating transistors are packed closer together. Thus, while the chip itself might not be as hot, its power density (the amount of heat concentrated on particular spots of the chip's surface) may begin to increase. Efficient cooling methods are required to protect spots in the chip from heating. If heat is not properly removed or managed, it will shorten the IC's overall life, even destroying the IC.

Heat buildup in an IC generally begins as junction temperature rises until heat finds a path to flow. Eventually, thermal equilibrium is reached during steady-state operating temperature, which affects mean time between failure (MTBF) of the device. A frequently used rule of thumb is that for each 10°C rise in junction temperature, there is a doubling of the failure rate for that component. Thus, lowering temperatures 10°C to 15°C can approximately double the lifespan of a device. Accordingly, designers must consider operating temperature as well as safety margin of devices.

Capacitors. Among discrete passive components, capacitors are the most sensitive to elevated temperatures. Lack of compact, thermally stable and high-energy-density capacitors has been one of the most significant barriers to the development of high-temperature systems. For traditional ceramic dielectric materials, there is a fundamental tradeoff between dielectric constant and temperature stability.

Capacitance of devices made with low-dielectric-constant titanates, such as COG or NP0, remains practically constant with temperature and shows little change with aging. Capacitance of devices made with high-dielectric-constant titanates, such as X7R, is larger but exhibits wide variations with increase in temperature. In addition, leakage currents become unacceptably high at elevated temperatures, making it difficult for the capacitor to hold a charge.

There are few alternatives. For example, standard polymer film capacitors are made of polyester and cannot be used at temperatures above 150°C because their mechani-

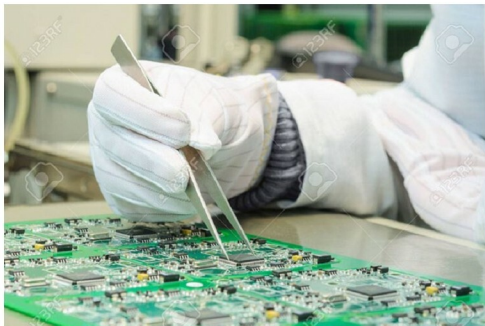


Fig. 1: Understanding the effects of uncontrolled environmental conditions at the component and system levels and applying this knowledge during the design phase can improve the reliability of the equipment

cal integrity and insulation resistance begin to break down.

Polymer films, such as PTFE, are mechanically and electrically stable at higher operating temperatures—showing minimal changes in dielectric constant and insulation resistance even after 1000 hours of exposure to 250°C. However, these films also have the lowest dielectric constant and are the most difficult to manufacture in very thin layers, which severely reduces the energy density of capacitors.

The best option is to use stacks of temperature-compensated ceramic capacitors. New ceramic dielectric materials continue to offer improved high-temperature stability via tailoring of the microstructure or the composition of barium-titanate-based mixtures. One particularly promising composition is X8R, which exhibits the energy density of X7R but has a minimal change in capacitance to 150°C.

Resistors. Since flow of electrons through any medium develops heat, all resistors are subject to temperature changes. The effect of these temperature variations depends upon the type of construction and is known as temperature coefficient.

Temperature coefficient indicates how temperature changes affect the resistor's value and may be either positive or negative. In general, com-

position resistors have a negative temperature coefficient, and metallic or wire-wound resistors have a positive coefficient. This means that composition resistors decrease in resistance with an increase in temperature, while metallic types increase in resistance with an increase in temperature.

A low temperature coefficient indicates slight change in resistance per degree of temperature change. High-quality resistors have a low or even zero temperature coefficient. These, of course, are the most desirable, especially in precision work.

Composition resistors are commonly found in electronic circuits. The element's resistive material is moulded into a small rod or deposited upon an insulating core. Wire leads are coaxially attached to each end of the element and an outside covering of natural bakelite is applied for insulation. The resistance value is marked on the body using EIA colour code.

Resistance values range from a fraction of an ohm to several million ohms. Exact values are difficult to manufacture and usually are not required. Therefore tolerance limits of 5 and 10 per cent are often used. In each tolerance group only certain preferred values are made. So no overlapping of values is possible due to normal manufacturing variation.